

Survey-Weighted Longitudinal Data Science for Preschool Shared Reading and Adolescent Vocabulary in Chile

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Abstract. Longitudinal education data require explicit treatment of sampling design, follow-up selection, and incomplete covariates. We examined associations of preschool shared reading and observed children’s books with adolescent receptive vocabulary in Chile’s Encuesta Longitudinal de Primera Infancia (ELPI), using children aged 36–71 months in 2012 who were reassessed 12 years later, in adolescence. The primary analysis used survey-weighted linear regression in a domain of 5,016 participants, adjusting for family characteristics and preschool vocabulary; eleven sensitivity specifications covered weighting, sample composition, caregiver and intermediate vocabulary, outcome scale, functional form, and missing data. Compared with never, shared reading at any frequency was associated with 1.74–2.17 additional standard-score points (daily: 2.17; 95% CI 0.83–3.51; Holm-adjusted $p = .006$), about 0.14 of the nominal reference standard deviation; estimates for the three frequencies were similar and differences among them were not tested. At least ten visible children’s books were associated with 0.94 points, with an interval including zero (−0.06 to 1.93). A separate exploratory, unweighted analysis used leave-one-region-out validation: ridge regression lowered RMSE from 15.32 to 14.04 relative to a training-mean benchmark (pooled out-of-region $R^2 = 0.16$), a modest gain. Associations were modest and do not establish causal effects, optimal reading frequency, or predictive performance for individual children.

Keywords: Longitudinal analysis · Survey regression · Shared reading
· Receptive vocabulary · Missing data · Geographic validation

1 Introduction

The Home Literacy Model distinguishes meaning-centered interactions from explicit literacy teaching [20,21]. Caregiver speech and storybook interaction quality matter beyond reading frequency [17,6]. Intervention meta-analyses support language benefits with variation across designs [5,15], but do not establish long-term effects of naturally occurring reading. Earlier syntheses had already linked

joint book reading with language growth, emergent literacy, and later reading achievement [28].

Early reading predicts later vocabulary [10], yet reciprocal child-environment processes [4] and caregiver language abilities [16] complicate interpretation. Earlier Chilean ELPI research addressed home learning, literacy practices, and vocabulary [23,27,1,25]. The 2024 follow-up extends observation into adolescence.

We jointly modeled mother/female-guardian reading and observed children’s books in 2012 against vocabulary in 2024. The contribution is an auditable longitudinal application distinguishing survey domains, sample composition, selection, and missing covariates. We estimate adjusted associations; predictive performance requires separate validation [22].

The question is also a data-science problem. Public microdata from several waves must be linked, weights must be matched to the panel that was actually observed, incomplete covariates must be handled explicitly, and any statement about prediction must be tested on data that were not used for fitting [22,34]. Design-based regression [8,36] targets population-level associations, whereas penalized (ridge) regression [29,32,33,37] offers a transparent way to quantify out-of-sample accuracy. We therefore report the survey-weighted association analysis as the primary result and add, as a clearly separated exploratory analysis, an unweighted description and a geographic out-of-region validation of the supplied cohort file. The prediction component is deliberately modest and is not presented as a machine-learning contribution.

2 Data and statistical methods

Table 1 states the status of each group of analyses. Only the survey-weighted M2 model is the primary analysis; the sensitivity analyses test its robustness, and the exploratory analyses are descriptive and use a different, unweighted cohort definition. The study was not preregistered, and the sensitivity plans were recorded after the initial reading results were available.

2.1 Cohort linkage and analytic population

ELPI began in 2010 with a national random sample of 15,175 children aged 0–4 years, who were followed in 2012, 2017, and 2024; the 2012 round added a refreshment sample, and the 2024 round interviewed 10,003 adolescents who had started the panel in 2010 [13,11]. Public ELPI microdata covered 2010, 2012, 2017, and 2024 [12,13,14,11]. The 2012 (second round) and 2024 (fourth round) data, with their questionnaires, user manuals, and original Stata files, were obtained from the Observatorio Social of the Ministerio de Desarrollo Social y Familia [13,11]. Original-cohort identifiers were checked for uniqueness; household rosters were reduced to relevant child/caregiver records. Exposures and preschool vocabulary came from 2012, family characteristics and caregiver vocabulary from 2010, intermediate vocabulary from 2017, and outcomes from 2024. Age-specific instruments were not substituted.

Table 1. Hierarchy and status of the analyses.

Level	Analyses	Status
Primary	Survey-weighted M2 model for the three reading contrasts and the ten-book indicator, with Holm adjustment	Defined first; not preregistered
Sensitivity	Eleven specifications (44 contrasts): weighting, sample composition, caregiver and 2017 vocabulary, outcome scale, splines, follow-up weighting, covariate imputation	Plans recorded after initial reading results
Secondary sensitivity	E-values for unmeasured confounding	Approximate for a continuous outcome; not evidence of an effect
Exploratory	Unweighted description of the supplementary cohort; leave-one-region-out ridge prediction	Different cohort definition; no population inference

Figure 1 shows sequential eligibility: 11,692 original-cohort participants, 9,561 aged 36–71 months, 9,423 with valid exposure/adult-book measures, 8,858 baseline-complete, 6,295 with the 2024 outcome, and 5,016 with four-wave weights. Unadministered forms were not imputed. The supplement documents coding, exclusions, and all exposure models.

2.2 Measures and adjustment variables

Reported mother/female-guardian reading or picture-book viewing in the previous seven days was categorized as never, 1–3 times, 4–6 times, or daily; unknown/not-applicable responses were missing. This item excludes other adults’ reading and interaction quality; respondents need not be the named adult. HOME2 recorded at least ten visible children’s books, an observation threshold rather than a household inventory. Adult books were a covariate; HOME1 did not replace HOME2.

The outcome was the 2024 Hispanic-American standardized score of the Test de Vocabulario en Imágenes Peabody (TVIP). Its historical reference scale has nominal mean 100 and standard deviation 15; it is not a contemporary Chilean norm [11]. Raw scores were a sensitivity outcome. Scores were not differenced across waves, and longitudinal measurement invariance was not assumed.

Family adjustment included recorded sex, region, urban/rural area, caregiver age and education, household size, and household income, measured in 2010. Education was basic or less, secondary, or tertiary; special education was excluded rather than merged with basic education. Nominal household income categories were up to CLP 250,000, CLP 250,001–450,000, and above CLP 450,000, retaining valid reported bands when amounts were unavailable. Child age in months and raw TVIP in 2012 entered with quadratic terms. No automated variable selection or interaction search was performed.

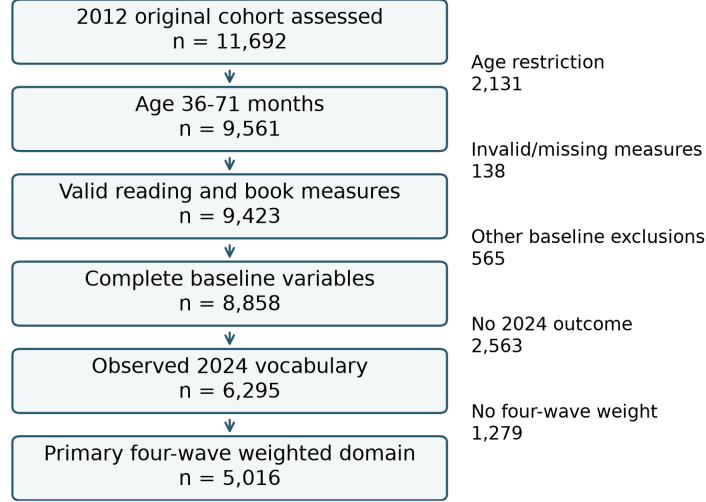


Fig. 1. Participant flow from the original 2012 cohort to the primary survey domain. Exclusions are sequential. The separate education/income imputation sensitivity includes 6,586 participants.

2.3 Primary analysis: estimand and survey estimation

The primary estimands were the three adjusted reading-category differences relative to never and the difference for observing at least ten children’s books. For child i , the primary conditional mean model M2 was

$$\begin{aligned}
 E(Y_i \mid R_i, B_i, A_i, V_i, C_i) = & \beta_0 + \sum_{k=1}^3 \beta_k \mathbb{I}(R_i = k) + \beta_B B_i \\
 & + \gamma_1 A_i + \gamma_2 A_i^2 + \gamma_3 V_i + \gamma_4 V_i^2 + C_i^T \delta.
 \end{aligned} \tag{1}$$

Here Y is the 2024 standard score, R is reading category (0 = never), B is the children’s-book indicator, A is age in months, V is raw 2012 TVIP, and C contains the family covariates and adult-book indicator. Reading was categorical, with no imposed ordinal trend. The additive specification estimates common adjusted differences across C . M1 omitted the V terms on identical participants, separating the change in adjustment from sample composition. M2 is the primary specification because the question is the association between early reading and adolescent vocabulary among children with similar preschool vocabulary and family background, not the total effect of reading; M1 is reported to show how much of the association is shared with concurrent preschool skill.

Survey weights were chosen by comparing each weight’s membership with the observed interview pattern (Table 2). A weight was accepted only if every child

Table 2. Audit of the ELPI weights available in the 2024 database. Waves are those implied by the variable name.

Weight	Waves	Positive	2012 int.	No 2012 int.	2012 int., no wt.	In suppl.
f_exp	2024	10,003	8,910	1,093	0	6,688
f_exp_1024	2010, 2024	10,003	8,910	1,093	0	6,688
f_exp_101224	2010, 2012, 2024	7,668	7,012	656	1,898	5,317
f_exp_101724	2010, 2017, 2024	8,910	8,910	0	0	6,688
f_exp_10121724	2010, 2012, 2017, 2024	7,012	7,012	0	1,898	5,317

Positive: children with a positive weight. 2012 int.: of these, children with a 2012 interview; No 2012 int.: without; 2012 int., no wt.: children interviewed in 2012 with no positive value; In suppl.: children of the supplementary cohort ($n = 6,688$) with a positive value.

with a positive value had the interview that its name implies and its positive values corresponded to the linked panel. The four-wave weight `f_exp_10121724` met both conditions: its 7,012 positive values were all among children interviewed in 2012, and the authors’ linkage identified the same children as observed in all four waves. The weight `f_exp_101224` did not meet the first condition, because 656 of its 7,668 positive values were for children without a 2012 interview; `f_exp_101724`, whose name omits 2012, was positive exactly for the 8,910 children interviewed in 2012. These two weights were therefore not used or relabeled. The provider’s weights report for the 2024 round describes how the factors were constructed [11].

The 7,012-person survey frame comprises all children with a positive `f_exp_10121724` weight, all of whom were interviewed in 2012, irrespective of age, exposure, or outcome availability. It was kept intact so that variance estimation treats the 5,016-person analytic sample as a subpopulation of the full design: the domain restricts this frame to children aged 36–71 months with valid exposures, complete baseline covariates, and an observed 2024 outcome. Following the provider’s adolescent-analysis guide, we specified individual sampling units, 116 strata, and the supplied four-wave weights. With D denoting the domain and x the full model vector, coefficients solve

$$\hat{\theta} = \arg \min_{\theta} \sum_{i \in D} w_i (Y_i - x_i^T \theta)^2. \quad (2)$$

Taylor linearization supplied the survey covariance matrix [8]; ordinary weighted-least-squares standard errors were not substituted. Individual confidence intervals used the fitted survey model’s residual degrees of freedom. A joint Wald F test assessed the three reading coefficients. Holm adjustment covered the four primary exposure tests [7]; displayed 95% intervals are pointwise and remained nominal. Sensitivities were not treated as independent replications. We did not test whether daily reading exceeded the other nonzero frequencies.

Concurrent preschool TVIP may reflect earlier reading, so M2 does not identify a total causal effect. Intermediate vocabulary adjustment can additionally remove experience-related pathways [18]. Neither adjustment nor follow-up duration eliminates familial confounding.

2.4 Sensitivity analyses and missing data

Eleven specifications produced 44 exposure contrasts. Unweighted M2 used HC3 heteroskedasticity-consistent covariance [9] for all 6,295 complete linked participants and separately for the exact 5,016-person survey domain. This distinguishes sample composition from weighting. Other survey specifications used raw 2024 vocabulary, added raw 2017 TVIP and its square ($n = 4,666$), or replaced quadratic age and preschool TVIP terms with natural cubic splines, three degrees of freedom each, using default knots in the analytic domain. The 2017 adjustment is not a mediation analysis.

Caregiver WAIS vocabulary in 2010 was available for 4,851 primary-domain participants. M2 and M2 plus linear and quadratic caregiver vocabulary were fitted to this identical subpopulation. Continuity of caregiver identity between 2010 and the adult referenced in 2012 was not established.

For follow-up selection, logistic regression predicted observation S of the 2024 outcome from all M2 predictors among 8,858 baseline-complete participants. Observed participants received stabilized weights

$$w_i^{\text{sel}} = \frac{\hat{P}(S = 1)}{\hat{P}(S_i = 1 \mid R_i, B_i, A_i, V_i, C_i)}, \quad S_i = 1. \quad (3)$$

Weights were truncated at their first and ninety-ninth percentiles among observed participants. Percentile confidence intervals came from 1,000 individual bootstrap resamples [35], refitting selection and outcome models and recalculating truncation within each resample (seed 20260923). This sample-based sensitivity requires adequate observation probabilities and conditional independence of missingness from the outcome given modeled predictors [19]. It is not a bootstrap of the complex survey design. Survey weights and selection weights address different problems, representativeness and attrition, and they were not multiplied: the four-wave weight exists only for the 7,012 children observed in all waves, whereas the selection model was fitted among all 8,858 baseline-complete children, and whether the provider’s four-wave weight already contains a non-response adjustment should be established from the weights report [11] before combining them. A sensitivity analysis combining design and attrition weights is therefore left as a limitation.

Missing education and income were addressed separately through substantive-model-compatible fully conditional specification [3]. Among 6,657 participants with observed exposures and outcome, 63 lacked preschool vocabulary and eight were excluded for special education, leaving 6,586; 291 lacked education or income. We used multinomial covariate models, 30 imputations, 30 iterations, the M2 substantive formula, and independent random-number streams initialized with seed 20260926. Exposures and vocabulary were not imputed. Each completed dataset used unweighted M2 with HC3 covariance. Rubin pooling [31] combined within- and between-imputation uncertainty, with finite complete-data degrees of freedom [2]. This sensitivity assumes conditionally missing-at-random covariates and adequate models [24,30]; it cannot recover loss to follow-up or restore survey representativeness,

and because it is unweighted its estimates are not directly comparable with the survey-weighted population estimand.

As a secondary sensitivity analysis, we computed E-values [38] for the primary contrasts. The adjusted mean difference was divided by the nominal reference standard deviation of 15 to obtain a standardized difference d , converted to an approximate risk ratio by $RR \approx \exp(0.91 d)$, and the E-value was $RR + \sqrt{RR(RR - 1)}$ for the point estimate and for the confidence limit closest to the null. This conversion is an approximation for a continuous outcome; E-values are descriptive and are not used as evidence for or against an effect.

The investigation was exploratory, not prospectively preregistered. It followed an earlier attendance analysis in the same data; sensitivity plans were recorded after initial reading results. R 4.5.2, survey 4.5, and smcfcs 2.0.2 supported estimation; Python supported linkage and selected independent numerical checks. Reporting followed STROBE principles [26]. All specifications remain available rather than retaining only favorable contrasts.

2.5 Exploratory analyses of the supplementary cohort

A separate analysis used the exploratory cohort file prepared from the 2012 and 2024 assessment files, the 2012 household roster, and the 2012 interview file [13,11]. It contains children aged 36–71 months in 2012 for whom both the 2012 and 2024 TVIP scores were available: 14,438 children were assessed in 2012, 9,669 met the age criterion, 6,766 had a 2024 assessment, and 6,688 had both scores. This definition does not apply the valid-exposure and baseline-completeness restrictions of the primary domain, so its counts differ from Figure 1; Table 5 sets the two flows side by side. No survey weights were applied. All 6,688 children have a positive 2024 weight (`f_exp`) and a positive `f_exp_101724` weight, which by its name covers 2010, 2017, and 2024 and is a different weight from the four-wave `f_exp_10121724` (Table 2); 1,371 of them lack the four-wave weight used in the primary analysis, for reasons that were not examined.

Variables were child age in months, supplied 2012 TVIP score, sex, maternal education code and household size from the 2012 roster, region, and seven binary home-observation items from the 2012 assessment (conversation with the child, verbal responses, praise, physical affection, games teaching colors or sizes, at least three puzzles, and at least ten children’s books). Missingness was summarized per variable. The reading-frequency item is not part of this file.

For prediction, the outcome was the 2024 standardized TVIP score and evaluation used leave-one-region-out (LORO) validation: each of the 15 regions was held out in turn, models were fitted on the other 14, and out-of-region predictions were pooled. Three predictors were compared: the training-fold mean; ridge regression [29] with basic child-level variables (age in months, 2012 TVIP, sex); and ridge regression with these plus maternal education code, household size, an indicator of a registered mother in the household, and the seven home-observation items. Within each training fold, missing values were imputed with the median, variables were standardized, and the ridge penalty was chosen by efficient leave-one-out cross-validation over 30 values between 0.01 and 1000 [32,33,34].

Table 3. Selected weighted characteristics of the analytic domain ($n = 5,016$).

Characteristic	Mean or %	95% CI
Child age in 2012, months	53.53	53.23 to 53.83
Child vocabulary in 2012, raw points	30.66	30.12 to 31.21
Vocabulary in 2024, standard-score points	108.26	107.82 to 108.70
Household size in 2010, persons	4.93	4.89 to 4.98
Female, %	48.86	47.40 to 50.33
Reading: never, %	23.51	22.29 to 24.73
Reading: 1–3 times, %	46.72	45.26 to 48.17
Reading: 4–6 times, %	13.43	12.45 to 14.41
Reading: every day, %	16.35	15.26 to 17.44
At least ten visible children’s books, %	30.63	29.29 to 31.97

Continuous entries are means; other entries are percentages. CI: nominal 95% confidence interval. Full descriptive results accompany the reproducibility files.

Accuracy was measured by mean absolute error (MAE), root mean squared error (RMSE), and $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2$, computed once over all pooled out-of-region predictions, so each child contributes equally. Because one region holds a third of the children, we also report macro-RMSE, the unweighted mean of the 15 regional RMSE values. Because children are clustered in regions, 95% intervals came from a percentile bootstrap that resampled regions (2,000 resamples, seed 42). An earlier single hold-out of regions 1, 10, and 12 ($n = 586$) from the original analysis script gave similar accuracy (RMSE 14.59, $R^2 = 0.17$), but the reason for choosing those regions is not documented, so it is not used for inference. Code for the LORO analysis accompanies the reproducibility files.

3 Results

3.1 Primary analysis

Follow-up outcomes were observed for 71.1% of baseline-complete participants. Of 9,561 age-eligible children, 703 lacked baseline analysis variables; overlapping missing counts were 92 for books, 46 for reading, 111 for preschool vocabulary, 99 for education, and 386 for income. Table 3 describes the restricted survey domain.

All three reading categories had positive adjusted differences relative to never (Table 4; Fig. 2). Daily reading was associated with 2.17 standard-score points (95% CI 0.83–3.51; Holm $p = .006$). The joint reading test was $F(3, 4871) = 4.18$, $p = .0058$. The three estimates (1.74, 1.82, and 2.17) were similar with overlapping intervals and we did not test differences among them, so the data support an association of shared reading at any frequency with vocabulary more than a graded benefit of daily reading. The children’s-book estimate was 0.94 points (−0.06 to 1.93; Holm $p = .066$); the interval includes zero, so the primary analysis does not show a clear book association. Adding preschool vocabulary

Table 4. Adjusted differences in adolescent receptive vocabulary ($n = 5,016$).

Exposure contrast	M1: difference (95% CI)	M2: difference (95% CI)	M2 Holm p
Reading 1–3 times	2.12 (1.00, 3.25)	1.74 (0.64, 2.83)	.006
Reading 4–6 times	2.08 (0.58, 3.58)	1.82 (0.39, 3.26)	.026
Reading every day	2.89 (1.50, 4.28)	2.17 (0.83, 3.51)	.006
At least ten visible children’s books	1.40 (0.36, 2.44)	0.94 (−0.06, 1.93)	.066

Reading reference: never; book reference: condition not observed. M1 adjusts for adult books, age and age squared, sex, region, area, education, income, caregiver age, and household size. M2 adds raw 2012 TVIP and its square. Confidence intervals are pointwise 95% intervals and are not multiplicity-adjusted; Holm p values cover the four M2 exposure tests.

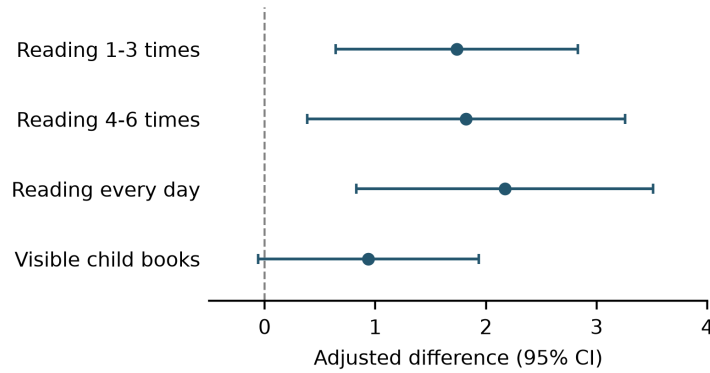


Fig. 2. Primary adjusted associations (M2; $n = 5,016$). Points and bars show differences and pointwise (not multiplicity-adjusted) 95% confidence intervals in TVIP standard-score points. Reading reference: never; books reference: condition not observed.

attenuated daily reading from 2.89 to 2.17 points and visible books from 1.40 to 0.94 points. This dependence on the adjustment set does not establish a causal mechanism or a dose-response relationship.

3.2 Sensitivity analyses

Figure 3 shows daily-reading estimates across ten standard-score specifications. The unweighted estimate was 1.97 points (0.83–3.10) in 6,295 participants and 2.13 (0.85–3.41) in the exact primary domain. The spline specification yielded 2.18 (0.84–3.53). Within the caregiver-vocabulary subpopulation, the estimate declined from 2.11 (0.74–3.48) to 1.86 (0.50–3.21) after WAIS adjustment. Adjustment for 2017 TVIP yielded 1.52 (0.27–2.78); the 4–6-times reading interval included zero. The raw-outcome estimate was 1.54 raw points (0.41–2.68), on a different scale.

Selection weighting yielded 2.01 standard-score points (bootstrap CI 0.92–3.20), with all 1,000 replicates valid. Estimated observation probabilities ranged

from .532 to .890 and truncated weights from .858 to 1.179. The largest absolute standardized difference across modeled terms between observed participants and the baseline-complete pool decreased from .040 to .0014. This describes follow-up balance, not exposure balance or elimination of confounding. The book interval included zero (-0.006 to 1.858).

After imputing education and income, daily reading was associated with 1.82 points (0.70–2.93) and visible books with 1.00 (0.15–1.86); these unweighted estimates are not directly comparable with the survey-weighted population estimand. There were no imputation warnings, and approximate coefficient Monte Carlo errors were below 0.4% of total standard errors for the four contrasts. These are numerical diagnostics, not tests of the missing-at-random assumption. Supplementary Table S2 reports all 44 estimates, including less consistent book associations and a flag for the unweighted specifications.

For the secondary E-value analysis, daily reading versus never (2.17 points) corresponds to $d = 0.145$, an approximate risk ratio of 1.14, and an E-value of 1.54; for the confidence limit closest to the null (0.83 points; $d = 0.055$, RR 1.05) the E-value was 1.28. An unmeasured confounder would need associations of about that size with both daily reading and vocabulary, beyond the measured covariates, to explain the estimate or to move the limit to the null. Because the book interval included zero, its confidence-limit E-value is 1. Such associations are plausible for familial factors such as caregiver language, so these values do not rule out confounding.

3.3 Exploratory description of the supplementary cohort

The unweighted exploratory cohort contained 6,688 children (mean age 53.6 months in 2012 and 15.9 years in 2024). Mean TVIP was 102.2 (SD 17.9) in 2012 and 108.1 (SD 15.3) in 2024, with a Pearson correlation of 0.32 between the two scores (Fig. 5). The 2024 distribution was left-skewed with a mode near 114 and a long lower tail (Fig. 4). Missing data were limited: maternal education was missing for 129 children (1.93%), the first home-observation item for 70 (1.05%), and the other six items for 63 each (0.94%); age, sex, household size, and both TVIP scores were complete. Table 5 relates this cohort to the primary analytic flow.

Children whose home visit recorded at least ten children’s books ($n = 1,960$) had a higher unadjusted mean 2024 TVIP score than those without ($n = 4,665$): 110.8 versus 106.9, with medians of 114 and 111 and standard deviations of 13.7 and 15.9 (Table 6; Fig. 5). Sixty-three children had no book item. Their 2012 scores also differed (105.8 versus 100.8), so the contrast is descriptive and confounded by baseline vocabulary and family circumstances.

Children in the exploratory cohort resembled those not retained. The 6,688 included children had mean age 53.6 months, mean 2012 TVIP 102.2, and 29.6% with ten books observed, compared with 2,981 excluded children (mean age 52.7 months, mean 2012 TVIP 101.3 among the 2,869 who had it, and 31.7% with ten books among 2,951 with the item) (Table 7).

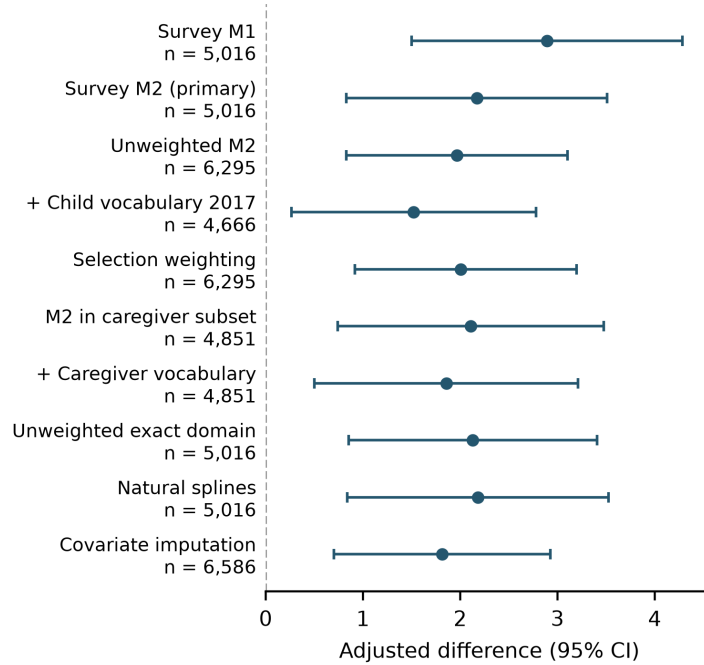


Fig. 3. Daily reading versus never across ten standard-score specifications. Bars are nominal 95% confidence intervals. Samples overlap; estimates are not independent replications. IPW intervals use 1,000 bootstrap resamples. The raw-score model is excluded because its units differ.

Table 5. Cohort flow in the primary analysis and in the supplementary cohort.

Stage	Primary analysis	Supplementary cohort
Assessed in 2012	11,692 (original cohort)	14,438 (all 2012 assessments)
Aged 36–71 months	9,561	9,669
Valid reading and book measures	9,423	not required
Baseline-complete	8,858	not required
With 2024 outcome	6,295	6,766 (6,688 with both TVIP scores)
Four-wave weight available	5,016	5,317 (of the 6,688)

The 2012 assessments include the refreshment sample of children born from 1 September 2009 to 31 December 2011 [13], most of whom were younger than 36 months in 2012 and none of whom was followed in 2024 [11]; this is consistent with the larger initial count (14,438 versus 11,692), and with the small difference in age-eligible counts (108 children), but it has not been verified at the folio level. The primary counts also apply restrictions on valid exposures and complete baseline covariates that cannot be reproduced from the supplementary file; a folio-level reconciliation script accompanies the reproducibility files.

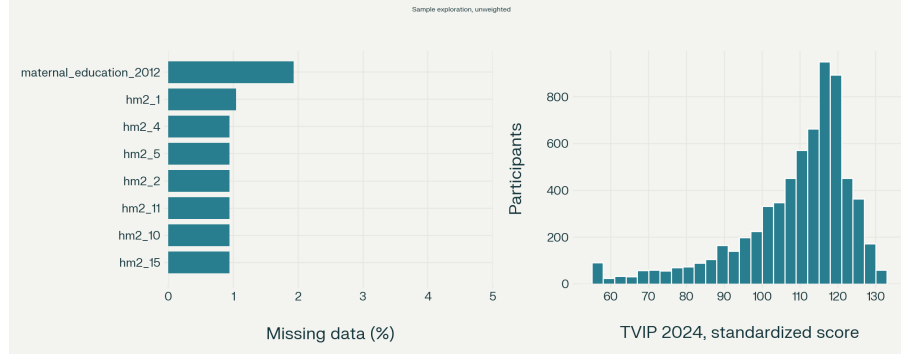


Fig. 4. Unweighted exploration of the supplementary cohort. Left: percentage of missing data in the variables with missing values. Right: distribution of the 2024 standardized TVIP score ($n = 6,688$).

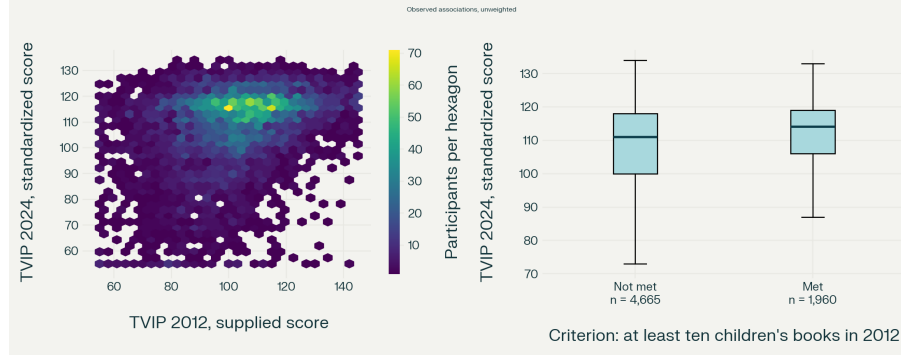


Fig. 5. Unadjusted, unweighted associations in the supplementary cohort. Left: hexagonal-bin density of 2012 versus 2024 TVIP scores ($r = 0.32$). Right: 2024 TVIP by the ten-book criterion (boxes show median and interquartile range; whiskers $1.5 \times$ IQR).

Table 6. Unadjusted 2024 TVIP by the ten-book criterion in 2012 (supplementary cohort, unweighted).

Ten books observed	n	Mean 2024	Median 2024	SD 2024	Mean 2012
No	4,665	106.94	111	15.86	100.76
Yes	1,960	110.77	114	13.66	105.85
Item missing	63	110.94	115	11.15	97.56

Descriptive statistics only; no adjustment or survey weighting.

Table 7. Children retained versus not retained in the supplementary cohort.

Group	<i>n</i>	Mean age (months)	Mean 2012 TVIP	Ten books (%)
Not retained	2,981	52.66	101.30	31.7
Retained	6,688	53.58	102.22	29.6

The 2012 TVIP was available for 2,869 not-retained children and the book item for 2,951 of them.

Table 8. Leave-one-region-out accuracy for the 2024 TVIP score ($n = 6,688$; 95% region-bootstrap intervals in parentheses).

Model	MAE	RMSE (pooled)	Macro-RMSE	R^2
Training-fold mean	11.58 (11.24, 11.93)	15.32 (14.76, 15.91)	15.09 (14.33, 15.93)	−0.002
Ridge: basic	10.67 (10.35, 11.12)	14.30 (13.82, 14.82)	14.08 (13.42, 14.78)	0.13 (0.10, 0.15)
Ridge: basic + household	10.48 (10.16, 10.93)	14.04 (13.57, 14.53)	13.87 (13.21, 14.55)	0.16 (0.12, 0.18)

Pooled out-of-region predictions. Macro-RMSE is the unweighted mean of the 15 regional RMSE values. Lower MAE and RMSE and higher R^2 indicate better accuracy.

3.4 Exploratory prediction by leave-one-region-out validation

Pooled out-of-region predictions for all 6,688 children are summarized in Table 8. The benchmark predicted each held-out region with the mean of the other 14 regions (15 distinct values between 107.9 and 108.3), giving RMSE 15.32 (95% region-bootstrap CI 14.76–15.91) and a pooled R^2 of −0.002. Ridge regression with basic variables reduced RMSE to 14.30 (13.82–14.82; $R^2 = 0.13$), and adding household and home-observation variables reduced it to 14.04 (13.57–14.53; $R^2 = 0.16$). The larger model lowered RMSE by 1.28 points relative to the benchmark (95% CI 1.02–1.53) and by 0.26 points relative to the basic model (0.18–0.34). It had lower RMSE than the benchmark in all 15 held-out regions (Fig. 6); region-level RMSE of the larger model ranged from 11.72 (region 11, $n = 63$) to 15.93 (regions 3 and 12), against 12.47 to 18.06 for the benchmark.

Region 13 contributes 2,302 of the 6,688 children (34%), so pooled metrics weight it heavily. With each region weighted equally, macro-RMSE was 15.09 (14.33–15.93) for the benchmark, 14.08 (13.42–14.78) for the basic model, and 13.87 (13.21–14.55) for the larger model, a gain of 1.22 points (0.90–1.56) over the benchmark, so the conclusion does not depend on region 13.

4 Discussion

Shared reading at any reported frequency was associated with higher adolescent vocabulary than no reading after family and preschool-skill adjustment, with similar point estimates across frequencies. The daily-reading difference was approximately 0.14 of the test’s nominal reference standard deviation, not the analytic sample standard deviation; educational importance is not established. Books showed less consistent evidence, and the primary estimate for at least ten visible books has an interval that includes zero.



Fig. 6. Out-of-region RMSE by held-out region (code and number of children). Grey: training-fold mean; teal: ridge with basic plus household variables. The ridge model had lower RMSE in all 15 regions.

Interactions and material resources represent distinct literacy dimensions [20,21]. Weekly frequency omits discussion, participation, language, and quality; visible books omit accessibility and use [17,6]. The exposure reflects one respondent’s report of the mother or female guardian over seven days, so misclassification of habitual reading is likely and may attenuate or otherwise distort the associations. This follow-up extends earlier ELPI findings [23,1], but subsequent reading and schooling remain unmeasured pathways.

Attenuation after caregiver-vocabulary adjustment supports caution about familial confounding [16] and reciprocal child-environment processes [4]. Household income was available only in three nominal 2010 bands, so residual socioeconomic confounding is also likely. The E-values (1.54 for the estimate and 1.28 for the confidence limit) are modest, so an omitted factor such as caregiver language ability could plausibly account for the association [38]. These associations do not isolate environmental causal effects or evaluate supported reading interventions [5,15]. They establish neither optimal frequency nor sole maternal responsibility; broader cultural literacy practices remain relevant [27].

Survey inference, selection weighting, and covariate imputation target different populations under different assumptions. Agreement cannot exclude bias. Because the sensitivity plans were recorded after the initial reading results, the sensitivity analyses should be read as checks on the primary estimate and not as independent replications, and a specification combining design weights with attrition weights is not provided. Exploration, one-week exposure measurement, missing baselines, and attrition limit inference; outcome-dependent missingness was not modeled. Historical norms and untested invariance preclude individual growth claims. Results cannot be directly generalized to Puno or Peru.

The exploratory prediction analysis illustrates the distinction between explanation and prediction [22]. Out-of-region, a ridge model explained about 16% of the variance in adolescent scores, and the household and home-observation

variables added only 0.26 RMSE points over basic variables. This is consistent with a large role of unmeasured factors and of the persistence captured by the earlier score ($r = 0.32$). The analysis is unweighted, uses a different cohort definition from the primary analysis (Table 5), excludes the reading-frequency item, relies on 15 regions of very unequal size, and uses a single model specification. It is not a validation of the survey-weighted estimates and should not be read as a machine-learning contribution. External or temporal validation would be needed before any individual-level use [22,34]; these associations are not a validated screening tool.

5 Conclusion

ELPI shared reading, at any reported frequency, was associated with modest adolescent vocabulary differences across sensitivity specifications; visible-book evidence was less consistent. Transparent survey, selection, and missing-data assumptions support interpretation without establishing causality or predictive accuracy. An exploratory out-of-region evaluation showed only modest accuracy, reinforcing that these results should not be used to classify individual children.

Data and computational transparency

Public microdata and technical reports are available from the ELPI portal of the Ministerio de Desarrollo Social y Familia, including the second round (2012) and fourth round (2024) pages with questionnaires, user manuals, and original data files [13,11], and the first and third rounds [12,14]. Accompanying reproducibility materials contain scripts and aggregate outputs, including the leave-one-region-out code, the weight audit, and a cohort reconciliation script; participant-level and imputed records are not redistributed. AI-assisted tools (OpenAI Codex) supported scripting, numerical cross-checking, and manuscript drafting. The methods and results are documented in the analysis code and aggregate outputs.

Declarations

Funding. This research received no external funding. **Competing interests.** The authors declare that they have no competing interests. **Ethics statement.** This study is a secondary analysis of publicly available, de-identified microdata from the Encuesta Longitudinal de Primera Infancia (ELPI), distributed by the Ministerio de Desarrollo Social y Familia of Chile. No new data were collected, no participants were contacted, and no personally identifying information was accessed; accordingly, no additional ethics review was undertaken. Data collection and informed-consent procedures were the responsibility of the data provider.

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